Universidad [Nombre de la Institución]
Department of Data Science and Artificial Intelligence

Project Proposal — Activity 2, Unit 2

Urban Sprawl Prediction in Mérida, Yucatán (2025–2030) Using Cellular Automata and Random Forest

Course:	Integrative Project II
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1. Project Title

Urban Sprawl Prediction in Mérida, Yucatán (2025–2030) Using Cellular Automata and Random Forest

2. Problem Statement / Justification

Mérida, Yucatán, has experienced one of the fastest urban growth rates in Mexico over the past decade, driven by industrial expansion, tourism, and migration. This rapid, often unplanned expansion — known as *urban sprawl* — generates significant environmental and socioeconomic consequences: loss of agricultural land and natural vegetation, increased urban heat island effects, strain on water and infrastructure networks, and growing inequality in service access.

Currently, municipal planning agencies lack predictive tools capable of identifying which specific areas are most at risk of urbanization in the short-to-medium term. Decisions are made reactively, after expansion has already occurred. This proposal addresses that gap by developing a spatially-explicit predictive model that combines satellite remote sensing data, machine learning, and cellular automata simulation to forecast urban growth at the cell level for the period 2025–2030.

Beneficiaries: Municipal urban planning offices, SEDATU, environmental agencies, researchers, and civil society organizations working on sustainable urban development in the Yucatán Peninsula.

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3. Hypothesis

If a spatiotemporal predictive model is developed that integrates LULC classifications from LANDSAT 8/9 satellite imagery, spatial variables (proximity to roads, urban edge, population density), and a Cellular Automata simulation calibrated with Random Forest transition probabilities, then it will be possible to predict urban expansion patterns in Mérida, Yucatán with a Figure of Merit ≥ 0.30 and Cohen's Kappa ≥ 0.70, generating annual probability maps for 2026–2030.

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4. Objectives

General Objective

To develop a predictive model for urban sprawl in Mérida, Yucatán for the period 2025–2030, combining Random Forest classification and Cellular Automata simulation applied to multitemporal satellite imagery, in order to generate annual probability maps that support evidence-based urban planning decisions.

Specific Objectives

SO1. Download and preprocess LANDSAT 8/9 Collection 2 imagery (2015, 2020, 2024) via Google Earth Engine to produce Land Use/Land Cover (LULC) classification maps for the Mérida metropolitan area.

SO2. Extract and calculate relevant spatial features (distance to roads, urban edge distance, NDVI, NDBI, slope, population density) from INEGI, SRTM, and GEE datasets.

SO3. Train and validate a Random Forest binary classifier to model urban/non-urban land transition probability using paired-year LULC change data.

SO4. Implement a Cellular Automata simulation module that integrates RF probabilities with neighborhood rules to generate annual urban extent projections from 2026 to 2030.

SO5. Evaluate model accuracy using the Figure of Merit (FOM), Cohen's Kappa, and AUC-ROC metrics, and produce final GeoTIFF and Shapefile outputs for each prediction year.

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5. Brief Theoretical Framework

5.1 Land Use / Land Cover Change (LULCC)

LULCC refers to the transformation of Earth's surface driven by human activities and natural processes. Satellite-based LULC mapping using indices such as NDVI (Normalized Difference Vegetation Index) and NDBI (Normalized Difference Built-up Index) derived from LANDSAT multispectral imagery is the standard approach for tracking urbanization at regional scale [1].

5.2 Random Forest for Urban Growth Modeling

Random Forest (RF) is an ensemble learning algorithm widely applied in geospatial classification tasks. In urban growth modeling, RF is used to learn the statistical relationship between observed land transitions and a set of spatial driver variables, outputting transition probability surfaces that quantify where non-urban cells are most likely to urbanize [2].

5.3 Cellular Automata (CA) Simulation

CA models represent land as a grid of discrete cells that change state according to transition rules and neighborhood conditions. Coupled CA–ML models — where machine learning provides transition probabilities and CA propagates them spatially over time — have become the dominant paradigm for urban sprawl simulation due to their ability to capture spatial autocorrelation and emergent growth patterns [3], [4].

5.4 Validation Metrics for LULC Change Models

The Figure of Merit (FOM) quantifies the proportion of observed change correctly predicted relative to all predicted and observed change, penalizing both false alarms and missed predictions. FOM values above 0.25 are considered acceptable for urban growth models. Cohen's Kappa and AUC-ROC complement FOM by measuring spatial agreement and classification discrimination power, respectively [5].

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6. Methodology

The project follows a five-stage pipeline implemented in Python:

Stage 1 — Data Acquisition: LANDSAT 8/9 Collection 2 Surface Reflectance images for 2015, 2020, and 2024 will be downloaded via Google Earth Engine (GEE) Python API. Municipal boundary and road network shapefiles will be obtained from INEGI Marco Geoestadístico 2023. DEM elevation data will be sourced from SRTM 30m via GEE. Population density at AGEB level will be obtained from INEGI Censo 2020.

Stage 2 — Preprocessing & LULC Classification: Imagery will be cloud-masked and composited. NDVI, NDBI, and EVI indices will be computed. A supervised binary classifier will produce LULC maps (Urban=1 / Non-Urban=0) for each reference year. Spatial variables will be computed using rasterio, numpy, and scipy (distance transforms, slope from DEM).

Stage 3 — Random Forest Training: LULC change pairs (2015→2020, 2020→2024) will be used to extract training samples. The RF model will be trained using scikit-learn with hyperparameter tuning via cross-validation. Feature importance analysis will be performed to identify the most influential spatial drivers.

Stage 4 — Cellular Automata Simulation: The calibrated RF transition probability surface will be integrated into a CA simulation module. The combined transition probability is: $P\_total(t) = \alpha \times P\_RF + \beta \times P\_neighbors + \gamma \times P\_stochastic$, where parameters are calibrated using grid search against observed 2024 urban extent. Five annual iterations will produce urban extent predictions for 2026–2030.

Stage 5 — Validation & Output Generation: Model accuracy will be assessed using FOM, Kappa, and AUC-ROC on held-out 2024 data. Final outputs will be GeoTIFF probability maps and Shapefiles for each prediction year, plus a metrics report.

Tools & Libraries: Python 3.11, Google Earth Engine Python API, scikit-learn, rasterio, geopandas, numpy, scipy, matplotlib, seaborn, pandas.

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7. Expected Results

• Annual urban extent probability GeoTIFF maps for Mérida (2026–2030) at 30m spatial resolution.

• Projected urban boundary Shapefiles identifying the areas most likely to be urbanized by 2030.

• A trained and serialized Random Forest model (rf_model.pkl) with documented performance metrics (target: FOM ≥ 0.30, Kappa ≥ 0.70).

• A quantitative analysis of the spatial drivers of urban growth in Mérida (feature importance ranking).

• A reusable, documented open-source Python codebase suitable for application to other Mexican cities.

The expected impact is a decision-support tool that enables municipal planners to anticipate high-risk urbanization zones and design proactive land-use policies before expansion occurs.

8. Activity Schedule (8 Weeks)

Week	Activity	Deliverable
1	Data acquisition: GEE imagery, INEGI shapefiles, DEM, census data	Raw dataset collection
2	Preprocessing: cloud masking, index computation, LULC classification	Classified LULC rasters (2015, 2020, 2024)
3	Spatial feature extraction: distance transforms, slope, population density	Feature raster stack
4	RF model training, cross-validation, hyperparameter tuning	Trained rf_model.pkl + performance report
5	CA simulation module development and parameter calibration	Calibrated CA module
6	Annual prediction runs (2026–2030), map generation	GeoTIFF & Shapefile outputs
7	Model validation (FOM, Kappa, AUC-ROC), results analysis	Validation metrics report
8	Final document writing, visualization, and project presentation	Final PDF report + slides

9. AI Use Declaration

For the preparation of this proposal, Claude (Anthropic) was used as a support tool to assist with structuring the document, improving academic writing clarity, and ensuring compliance with the assignment rubric. The technical content — project concept, methodology, model architecture, variable selection, and codebase — was developed, reviewed, validated, and refined entirely by the team members based on their own prior work (see: github.com/Yarroth/urban_sprawl_merida).

10. Bibliographic References

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